Shadow Mirror: A Consent-Gated Architecture for Reflective Human–AI Interaction

Executable Invariants, Refusal as a Safety Mechanism, and the Threat Confound

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Abstract

Systems that infer what a person "really" wants from the gap between what they say and what they do face a hazard that is rarely named: the same surface contradiction is produced by a hidden preference and by a threat response. Telling someone they secretly want something they are afraid of is a specific, foreseeable harm, and no published affect-aware architecture reports its rate.

This paper unifies two strands of the Shadow Mirror programme. The first is architectural. Reference architectures for human–AI interaction state their commitments — traceability, provenance, human authority — in prose, where a reader cannot tell whether an implementation keeps them. A property that cannot be checked cannot be violated *detectably*, and a commitment that cannot be violated detectably is not a commitment. We specify **eleven architectural invariants as decidable properties with a supplied decision procedure**: each is a callable check returning a verdict and the observation that produced it. The second strand is ethical. We make **refusal a first-class output**, hold contradiction open rather than resolving it, and represent a threat response as a competing hypothesis rather than as noise.

The two strands are one system because the safety result depends on the structural commitments. Shadow Mirror misreads a threat response as a hidden desire in **0 of 800 simulated episodes** (95% upper bound 0.37% by the rule of three), against **42.9%** for a stated-versus-behaviour mismatch heuristic and **75.1%** for a baseline admitting physiological arousal into the action channel. That result rests on representing a competing hypothesis rather than filtering, and on distinguishing an absent channel from a zero reading — both properties of the graph substrate, not of a classifier.

We are explicit about what the architecture does not buy. On latent-state recovery it is statistically indistinguishable from a median split across all 20 tested conditions. A supervised four-channel classifier, given ground-truth labels our system never sees, beats us at low channel separation — 5.0% against our 29.7%. Those labels do not exist in deployment, which makes the comparison an upper bound on what any supervised method could reach rather than an available alternative. We report both because a reviewer would find them otherwise.

This system is a research instrument. It is not a diagnostic device and is not suitable for clinical, educational, employment, credit, insurance or policing use.

Keywords: reference architecture, executable specification, architectural invariants, refusal, abstention, consent, affective inference, threat response, provenance, belief revision, EU AI Act.

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1. Introduction

Architecture papers make promises. *Every conclusion traces to evidence. Provenance is preserved. The system does not silently replace human judgment.* These are good promises, and they are why reference architectures are worth writing. They are also, almost always, unfalsifiable as stated. A reader cannot determine whether an implementation keeps them, an author cannot determine whether a refactor broke them, and a reviewer cannot distinguish an architecture that enforces them from one that describes them.

When the architecture is intended to support a person reasoning about themselves, that gap stops being academic. A large class of deployed systems infers preference from behavioural inconsistency. Recommender systems treat repeated engagement as revealed preference regardless of stated intent. Implicit-measure tools treat reaction-time asymmetries as leanings a person will not admit. Affect-aware interfaces treat physiological arousal as intensity of wanting. In each case the inference has the same structure: *you say one thing, you do another, therefore the second is the truth about you.*

That inference has a failure mode with an established clinical and evolutionary literature and no name at all in machine learning. Approach under stated reluctance is produced by hidden preference. It is also produced by threat — by the specific pattern in which a person moves toward something because moving away is unsafe. Arousal is produced by excitement. It is also produced by fear. The two are not distinguishable from the surface signal, and the cost of confusing them is asymmetric. Reading a hidden preference as a threat response wastes an opportunity. Reading a threat response as a hidden preference tells a person that their fear was a want — which is, precisely, what a person recovering from coercion has most often already been told.

We began from an architecture in the usual condition: invariants specified in prose, with a prototype whose relationship to them was undocumented. Testing the prototype against its own invariants found that **provenance preservation was violated** — evidence derived from two sources reported one parent and silently discarded the other — and that belief revision was **order-dependent to the tune of 95 percentage points**. Both defects were present in a document that asserted the invariants held.

Contributions.

1. **Eleven architectural invariants as decidable properties**, with a supplied decision procedure, a randomised property harness, and a negative test per failure mode (§5).
2. **A reflective graph substrate** with multi-parent provenance closure, contradiction as a first-class edge, transactional updates, and no third-party dependencies (§4.1).
3. **An order-invariant, exactly retractable belief operator**, with its axioms stated explicitly and enforced by test (§4.2).
4. **A measured harm rate** for the threat confound, with two baselines reflecting current practice, and an ablation showing the conditions under which the protection fails (§6).
5. **Consent as a mechanism rather than a policy**: a hash-chained append-only ledger that answers what was granted at a past instant, and a deployment gate that refuses the biometric stream in institutional contexts (§7).
6. **Two negative results reported as such**: the architecture buys no accuracy, and a supervised baseline beats our guard where labels exist (§8).

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2. Conceptual foundations: the harm, stated precisely

Consider a person who states that they do not want something. Five mechanisms can produce that statement together with the behaviour that follows. Three are genuine preferences. Two are not.

Mechanism	Stated	Behaviour	Arousal	Avoidance	Is it a preference?
Explicit aversion	low	withdraws	low	low	yes — they do not want it
Latent approach	low	approaches	low	low	yes — they want it and have not said so
Latent approach, aroused	low	approaches	**high**	low	yes — excitement
Threat response	low	withdraws	**high**	**high**	**no** — fear
Threat compliance	low	**approaches**	**high**	moderate	**no** — complies under threat

The fifth row is the dangerous one. **Threat compliance and latent approach are indistinguishable on the stated and behavioural channels alone.** Both show stated aversion and approach behaviour. A system reading only those two channels — which describes most deployed preference inference — cannot tell them apart, and will call the compliance case a hidden want.

The third and fourth rows are the second trap. Excitement and fear both raise arousal. A system that treats arousal as intensity-of-wanting will call the threat case a strong want.

Our target metric is the rate at which a system outputs `latent_approach` — *"you want this even though you say you don't"* — when the true generating mechanism was a threat process. We call it the **false hidden-desire rate**.

2.1 Why this is an architectural problem rather than a modelling one

The natural response is a better classifier. That response fails for a reason worth stating: the distinguishing evidence is frequently *not collected*, and a model has no way to represent the difference between a channel that read zero and a channel that was never read. Imputing a missing avoidance measure to zero is imputing *evidence of safety*. The fix is structural — an absent channel must be representable as absent, and a competing hypothesis must be representable as competing rather than filtered away — which is why this paper is about an architecture and not about a model.

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3. Research objectives

The programme asks four questions, and answers three of them.

	Question	Status
O1	Can architectural commitments about reasoning be made decidable, with a decision procedure a third party can run?	**Answered** (§5). Eleven invariants, 200-seed randomised harness, negative test per failure mode
O2	What is the rate at which a reflective system misreads a threat response as a hidden desire, and does refusal remove it?	**Answered in simulation** (§6). 0 of 800 episodes; two baselines at 42.9% and 75.1%
O3	Can consent, and the prohibitions a system declares, be enforced by mechanism rather than stated as policy?	**Partly answered** (§7). Six of twelve prohibitions machine-enforced
O4	Does any of this change how a person reasons about themselves?	**Not answered.** No human-subjects data exists. See §11 and Appendix B

O4 is the one that matters most for the system's stated purpose, and it is the one with no evidence. That asymmetry is deliberate: nothing in this paper is a statement about people.

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4. Architecture

4.1 The Unified Reflective Graph

Typed nodes — Observation, Evidence, Context, Hypothesis, Belief, Inquiry, Explanation, Knowledge, Limitation — and typed edges — provenance, supports, contradicts, contextualises, revises, targets, explains, limits. Untyped insertions are rejected: an untyped graph cannot support a traceability invariant, because there is nothing to quantify over.

Provenance is a DAG. `ancestors()` returns the transitive closure over *all* provenance parents. The prototype followed the first predecessor only, recovering 0 of 2 parents from a two-parent derivation. The formalism licensed it: the evidence object was specified with provenance as a scalar attribute. Writing it as a set is the fix, and the defect is a case study in formal notation shaping implementation error — the provenance literature had the multi-parent answer three decades earlier.

Contradiction is an edge, not an overwrite. `contested_hypotheses()` returns hypotheses carrying both supporting and contradicting evidence, so an explanation can report a tension rather than preferring the later write.

Updates are transactional. A `transaction()` context manager snapshots and rolls back, so a failure mid-update cannot leave a hypothesis whose belief moved but whose provenance edge was never written.

No third-party dependencies. The graph is an adjacency structure in the standard library. This is not minimalism for its own sake: the architecture's claim is local-first auditability, and importing a graph framework for a dictionary of lists would trade that for nothing. NetworkX was removed for this reason.

4.2 Belief revision

Leaving the belief state's type open — "probability distributions, weighted graphs, logical structures" — and then asserting properties of it is a common and serious error. Properties cannot be asserted of an object whose type is open. We state the axioms instead:

- **B1 Plurality** — assigns a value to every hypothesis simultaneously; no distinguished maximiser.
- **B2 Commutativity** — order of conditionally independent evidence does not matter.
- **B3 Invertibility** — evidence can be retracted exactly.
- **B4 Audit** — every application records prior, posterior, trigger, method, time.

B3 is not a convenience. **A system that cannot undo an inference cannot honour a withdrawal of consent**, which turns an ethical commitment into an architectural requirement.

Belief accumulates in log-odds keyed by evidence identity, which yields B2 (addition commutes), B3 (subtract the term) and idempotence under resubmission. The prototype clamped the posterior to [0.01, 0.99] *in state* after each update. Clamping is not invertible: once pinned, the information about how far past the bound the evidence pushed is destroyed. Five observations in three orders produced final beliefs of **0.0396, 0.9900 and 0.9900**.

The current package clamps only at display. Re-running all **120 permutations** of a five-observation sequence in this pass produced **a single distinct posterior**, with a maximum spread across permutations of **0.0** — exact agreement, not agreement to tolerance (`verify_claims.py` §2).

The independence assumption, stated. Sequential updating requires conditional independence. In reflective domains this is frequently false: two self-reports on consecutive days are correlated through mood, framing and recall, not only through the latent state. Keying contributions by evidence identity prevents resubmission compounding, and temporal decay down-weights clustering, but neither substitutes for a correlated-likelihood model. We state the assumption rather than rely on it silently.

4.3 Inquiry planning

Candidate questions are scored by expected information gain over the current belief, normalised by cost. The heuristic this replaces scored inquiries as one minus the absolute belief gap — a quantity that does not depend on the candidate observation at all, and therefore carries no information about what to ask.

Normalised regret against an information-theoretic oracle, 500 trials per condition:

Planner	Hypotheses close (gap 0.04)	Hypotheses separated (gap 0.40)
Expected information gain	**0.161 [0.138, 0.187]**	**0.195 [0.168, 0.223]**
Belief-gap heuristic	0.579 [0.547, 0.612]	**1.000 [1.000, 1.000]**
Random selection	0.550 [0.515, 0.584]	0.562 [0.528, 0.597]

Two behavioural points matter more than the numbers. The heuristic is **indistinguishable from random** when it fires, and it emits nothing at all once hypotheses are separated — which is backwards, because a confidently held wrong hypothesis is exactly when a discriminating observation is worth most.

4.4 Temporal modelling

Observations are timestamped and decay in influence, so that outdated evidence does not dominate. The decay is not cosmetic: with a 30-day half-life, decade-old evidence produces an inference in **0.0%** of trials — the system declines rather than reporting a confident conclusion drawn from a person who no longer exists in the data.

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5. Invariants as an executable specification

5.1 The eleven invariants

	Invariant	Operationalisation
I	Evidence traceability	no explanation node lacks a reachable evidence node
II	Provenance preservation	every derived evidence node recovers **all** parents under transitive closure
III	Revision history	every belief transition records prior, posterior, triggering evidence, method, time
IV	Explicit uncertainty	no explanation is emitted without a non-empty uncertainty field
V	Hypothesis independence	contradicting one hypothesis removes or merges no competitor
VI	Context consistency	superseding a context annotates every dependent conclusion as stale
VII	Explanation integrity	explanations carry summary, uncertainty and assumptions
VIII	Inquiry justification	every inquiry quantifies expected gain and names the hypotheses it separates
IX	Transactional consistency	a failed update leaves no partial state
X	Human authority	no output phrased as an identity claim or directive; each carries uncertainty, a disclaimer and a question

	Invariant	Operationalisation
XI	Declared-only personhood	no attribute about a person exists unless they declared it or self-completed the instrument it came from; and no route exists to a global rank, a prescribed deficit, or an attributed purpose

Two required judgement rather than transcription.

VI cannot mean "recompute the past." An instrument cannot re-run a person's week. We operationalise it as annotation: when a context is superseded, every conclusion that depended on it is marked stale — neither silently retained nor silently deleted. Retention is the failure mode that produces confident conclusions drawn under conditions that no longer hold.

X is a property of what reaches the person, so it is checked at the output boundary rather than in the reasoning. Each terminal output must carry its uncertainty and disclaimer, offer at least one question, and avoid identity claims (*you are…*, *you secretly…*) and directives (*you must…*).

Enumerating the checks in the installed package in this pass returns **eleven** invariant checks plus an aggregate `check_all` (`verify_claims.py` §3).

5.2 The harness, and the negative tests

`build_well_formed(seed)` constructs a randomised reflective state: observations, derived evidence with multiple provenance parents, hypotheses with mixed supporting and contradicting edges, executed belief updates, planned inquiries, and an explanation. Invariants I–X are asserted across 200 seeds.

Invariant XI needed a second generator, because it ranges over a formation model rather than a graph. It is asserted across 200 randomised models, with 100 further randomised checks on two derived properties. Most instrument administrations in those models *refuse*, and that is the point: XI must hold whatever the model has been through, including the paths where every administration was declined.

Fourteen negative tests construct states that violate each invariant and assert the check catches them, alongside **four positive controls** — states that should pass and do. Earlier drafts reported "sixteen negative tests", which counted the positive controls among them; the correction is made explicitly rather than silently, because a paper about checkable claims should not miscount its own checks. Without these, a suite reporting "11/11 invariants hold" is unfalsifiable in exactly the way the prose was.

5.3 What happened when we pointed it at real output

The Invariant X check failed on its first contact with the shipped reflection layer. It flagged:

> *"You have consulted this 5 times. Seeing a guess changes later answers and can become self-fulfilling."*

as a verdict, on the pattern `you have`. That sentence is a factual statement about the person's own logged behaviour, and it is a caution the architecture specifically wants surfaced. **The defect was in the check.** We narrowed the patterns to identity claims and directives and recorded the reasoning in source.

We report this because it is the argument for the method. The check was written against a specification, passed against fixtures, and was wrong. It became correct only when aimed at output a person would actually see. A conformance suite that is never pointed at real output will pass, and will be worthless.

5.4 Do the checks discriminate between implementations?

A specification that only its own author's implementation can satisfy has demonstrated nothing.

The same decision procedure was run unmodified against an independently written implementation of the same architecture — a NetworkX-based prototype produced separately from this work. It passes four checks and **fails three**: provenance preservation (0 of 2 parents recovered), order invariance (spread 0.9504 across permutations), and inquiry justification (no information-gain content).

	Prototype	Current package
I Evidence traceability	pass	pass
II Provenance preservation	**fail** — 0/2 parents recovered	pass — 2/2
III Revision history	pass	pass
IV Explicit uncertainty	pass	pass
V Hypothesis independence	pass	pass — contradiction retained
VIII Inquiry justification	**fail** — no information-gain content	pass
Order invariance (120 permutations)	**fail** — spread 0.9504	pass — spread 0.0
	4/7	**7/7**

The procedure separates two implementations of one architecture, which is the minimum evidence that it measures conformance rather than restating an API. This is two implementations, not many, and §11 records that limitation.

5.5 An invariant that exists because of a request

Invariant XI was not derived from the architecture. It was derived from being asked for something.

A user of this system asked it to "determine and understand who people probably are according to their will, desire and need — where they come from, how mature they are, what they still need to be mature enough, what their eventual purpose probably is", with the qualification *"I would not say judge people."* That qualification is the interesting part: the request came with an intuition that something in it was wrong, and no way to say which part.

It is five requests, and they separate cleanly on whether a validated construct exists and who supplies it.

Ask	Disposition	Ground
will, desire, need	**built**	validated self-report exists; the person supplies it
where they come from	**declared only**	inference of special-category data; on biometric input, prohibited by AI Act Art. 5(1)(g)
how mature they are	**refused as a rank**	the rank *is* the construct, not a summary of it
what they still need	**refused**	a directive about identity; Invariant X already forbids it
eventual purpose	**refused**	no validated construct exists; any output would be invented

The maturity row took the most work. The instruments that measure developmental position — Loevinger's Total Protocol Rating, Kegan's subject-object position, Cook-Greuter's Center of Gravity — each yield a single ordered position in which the rank *is* the theory's claim, so a profile drawn from one would misrepresent it. We dropped the construct rather than softening it, and substituted instruments that never claim a stage.

What makes this architectural rather than a policy note is that a refusal stated in prose decays exactly as the original invariants did. Condition (b) of the Invariant XI check does not inspect source code or match names; it **calls** each method that ought to refuse and requires a typed refusal carrying the expected ground. Two failure modes are caught. A method that returns a value fails, obviously. A method that has been *deleted* also fails — and that is the less obvious commitment, because **absence is not refusal**. A caller who finds no `eventual_purpose` attribute learns nothing; a caller who calls it and reads why it declined learns what the architecture will not do and on what ground. The refusing methods therefore exist in order to refuse, which is a deliberate inversion of the usual advice to delete dead code.

Exercising these in this pass, nine refusing methods were called and **all nine raised a typed refusal with a stated ground; none returned a value** (`verify_claims.py` §6). What replaces the refused asks is not silence: `open_questions()` returns what the model does not know, phrased as invitations, and is guarded by a test that fails if the phrasing drifts into *lack*, *deficit*, *should* or *must*.

What this does not establish. A catalogue of refusal grounds is not a theory of which requests a reflective system should decline, and we do not have one. The check verifies that the named refusals refuse; it cannot verify that the catalogue is complete, and a request we did not anticipate will not be caught.

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6. Refusal as a safety mechanism

6.1 Four design commitments

C1. Refusal is a first-class output. The system emits `unresolved_approach` — a named state, not a low-confidence label — when behaviour approaches under stated reluctance and the evidence needed to exclude a threat response was not collected. It also refuses entirely below a minimum of paired evidence.

C2. Absent is not zero. An unobserved channel is recorded as unobserved, never imputed to a value, because imputing arousal or avoidance to zero is imputing *evidence of safety*.

C3. Threat is a competing hypothesis, not a filter. `threat_response` occupies the same hypothesis space as the preference readings and is scored against them, rather than applied as a post-hoc suppression rule.

C4. Contradiction is held, not resolved. Supporting and contradicting evidence are both retained as typed edges; a hypothesis carrying both is reported as contested rather than collapsed.

The output is a distribution over named states with an explicit uncertainty, or a refusal. There is no scalar score anywhere in the API.

How this differs from abstention as the literature defines it. Selective classification, learning to defer and confidence thresholding all abstain because a *confidence* is low — the model is unsure, so it declines, and the threshold is a knob trading coverage against accuracy. Our guard abstains because a *specific alternative explanation cannot be ruled out with the evidence that was collected*. It is not tunable by a confidence parameter, it does not move when the model becomes more certain, and it fires just as readily on a high-confidence reading. A selective classifier asks "how sure am I?"; the guard asks "which question was never asked?"

6.2 Method

Five generative processes, defined by target values on four channels (stated, action, arousal, avoidance), with observations drawn from Beta distributions around those targets. Ground truth is the generating process, which exists independently of the inference model.

Misspecification control. The inference model assumes each latent state sits at a fixed corner of a signal space. Generating from exactly those corners would test self-consistency rather than validity, so every generating centre is displaced by δ in a random direction.

Baselines, both implemented and executed at benchmark time. *Mismatch-only* uses stated versus behaviour with no confound channels, representing systems that infer preference from behavioural inconsistency alone. *Arousal-as-desire* folds physiological arousal into the action channel; this is not a straw man, as treating autonomic arousal as an approach signal is documented practice.

n = 400 episodes per process per condition, 6 observation rounds each, all rates carrying 2,000-resample percentile bootstrap intervals. The suite is deterministic: two full runs produce byte-identical output.

What is not measured. No human participants were involved at any stage. Every result below is a statement about simulated signal streams with known generating mechanisms, and none is a statement about people.

6.3 The safety result

δ = 0.10, both confound channels observed, n = 400 per process:

Method	False hidden-desire rate	True hidden preference detected
Shadow Mirror	**0.0% [0.0, 0.0]**	87.1% [84.9, 89.4]
Mismatch-only baseline	42.9% [39.5, 46.2]	100.0% [100.0, 100.0]
Arousal-as-desire baseline	75.1% [72.1, 78.1]	89.5% [87.4, 91.8]

Guard sensitivity 100.0% [100.0, 100.0]; specificity 94.4% [93.2, 95.7]. At δ = 0.00 the three rates are 0.0% [0.0, 0.0], 47.4% [44.0, 50.9] and 75.8% [72.8, 78.6], with specificity 97.4% [96.5, 98.3].

Per process, δ = 0.10:

True process	Shadow Mirror	Mismatch-only	Arousal-as-desire
`threat_response`	**0.0%**	0.0%	50.5%
`threat_compliance`	**0.0%**	**85.8%**	99.8%

The compliance row is the finding. A system reading stated-versus-behaviour alone gets the withdrawal-type threat response right — it looks like aversion — and gets the *compliance* case wrong **85.8%** of the time. That is the case where a person approaches because approaching is safer, and it is the case where being told "you actually want this" does the most damage.

The cost, stated plainly. Sensitivity to genuine hidden preference is 87.1% against the baseline's 100.0%. The entire loss is in the high-arousal approach case, detected 74.2% [70.0, 78.5] of the time, where excitement is mistaken for fear. Roughly 13 points of sensitivity buys the elimination of the misreading. We regard that as the correct side of the trade for an instrument that hands its output to a person about themselves, but it is a trade and not a free lunch.

6.4 The ablation: what "absent is not zero" is worth

Deployments frequently lack an avoidance measure. We ablate that channel and compare two treatments of the missing data.

	Imputes missing channel as zero	Records it as absent (ours)
Guard sensitivity	**0.0% [0.0, 0.0]**	**100.0% [100.0, 100.0]**
False hidden-desire rate	**31.2% [28.1, 34.4]**	**0.0% [0.0, 0.0]**
…on `threat_compliance`	62.5% [57.8, 67.2]	0.0% [0.0, 0.0]
Specificity	100.0%	67.0% [64.4, 69.8]
True hidden preference detected	99.2%	0.0% — the system refuses
AUC, fear vs excitement	—	0.835

Imputing a missing channel to zero silently disables the guard: sensitivity falls to zero and the compliance case is misread 62.5% of the time, with nothing in the output indicating that the protection is inactive.

Recording it as absent produces the opposite failure and the better one. The system stops asserting hidden preference entirely, and specificity falls to 67.0% because arousal alone cannot separate fear from excitement, so the caution fires on excitement too.

This is the central design claim. Under partial observation a reflective system should refuse, not degrade; and the refusal must be visible in the output, not implicit in a number that happens to be low.

6.5 Refusal calibration

Refusal is only defensible if it is calibrated. n = 300 per condition:

Condition	Behaviour
Evidence genuinely insufficient	refuses **100.0% [100.0, 100.0]**
Evidence sufficient and clear	over-refuses **0.0% [0.0, 0.0]**
Evidence ample but ambiguous	does not refuse; reports uncertainty **0.405 [0.391, 0.419]** against **0.021 [0.020, 0.021]** on clear evidence
Evidence present but expired (decade-old, 30-day half-life)	infers **0.0% [0.0, 0.0]** of the time

The system refuses when it should, does not refuse when it should not, and separates ambiguous from clear evidence by a factor of nineteen in reported uncertainty.

6.6 Ambivalence is a state, not a residual

"You want and do not want this at once" is a claim about a person. Computed from dispersion alone it fires on anything spread out: **95.5%** of pure measurement noise and **98.8%** of monotonic drift — that is, of someone who has *made up their mind* — read as ambivalence, for a precision of **34.0%**. Gating on lag-1 autocorrelation, which distinguishes alternation (−0.844) from drift (+0.517) and noise (−0.120) where dispersion cannot (0.357 / 0.245 / 0.257), raises precision to **97.6%** with recall unchanged at 100.0%.

6.7 Adversarial robustness: two checks a sceptical reader should demand

"Your simulator is your theory." The generative processes were defined so that threat and approach differ on arousal and avoidance, and the guard keys on arousal and avoidance. A 0.0% rate under those conditions risks being a tautology. So we swept the *separation* between the two processes on the confound channels, at n = 300 per process per level.

Separation	AUC	Shadow Mirror	Four-channel classifier	Mismatch-only
1.00	1.000	**0.0%** [0.0, 0.0]	**0.0%** [0.0, 0.0]	94.3% [91.7, 96.7]
0.75	1.000	**0.0%** [0.0, 0.0]	**0.0%** [0.0, 0.0]	95.3% [92.7, 97.7]
0.50	1.000	2.3% [0.7, 4.0]	**0.0%** [0.0, 0.0]	96.7% [94.7, 98.7]
0.25	0.988	29.7% [24.7, 35.0]	**5.0%** [2.7, 7.7]	97.3% [95.3, 99.0]
0.00	0.491	76.7% [71.7, 81.3]	82.0% [77.7, 86.3]	96.3% [94.0, 98.3]

The headline is not a knife-edge: it survives to roughly half the original channel separation. But it is **entirely contingent on the confound channels carrying signal**, and at zero separation every method fails, as it must — the classes are then identical on every observable, and any method reporting 0.0% would be reporting a bug.

"Your baselines are straw men." They partly were. The table therefore adds a baseline a competent engineer would build: a threshold classifier over the same four channels, with its threshold **fitted on a**

held-out labelled split. It matches us at high separation and **beats us at separation 0.25 — 5.0% against our 29.7%.**

We report this because it is true and because the reason matters: our guard uses a fixed, pre-specified threshold, while the classifier was handed ground-truth labels for "was this a threat response". **Those labels do not exist in deployment.** Nobody can tell a running system, for a given person and episode, which mechanism produced their behaviour; that is the entire difficulty. The supervised baseline is therefore an *upper bound on what any labelled method could achieve*, not an available alternative — and the gap at separation 0.25 is a real, quantified cost of not having supervision, which we would rather state than omit.

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7. Consent, governance and data ownership

7.1 Consent as a mechanism

The system's other-person mode requires granular, per-stream, purpose-locked, revocable consent before any datum is recorded. Streams a subject cannot notice about themselves — implicit measures, physiological arousal, micro-avoidance — cannot be granted under a blanket authorisation; each requires its own acknowledgement. Every stream declares in plain language both what it records and what it *may reveal*. No grant, no observation: the engine raises rather than records.

Three critiques bound what any such mechanism can claim, and we do not claim past them. Privacy is governed by norms of information flow appropriate to a context, so consent obtained in one context does not license a flow into another. Privacy self-management does not scale. Post-GDPR consent interfaces are routinely engineered to produce agreement rather than to elicit it. We therefore report an **enforcement ratio** rather than a consent rate: the question is not how many people agreed but how many refusals the mechanism actually produces.

7.2 The consent ledger

A mutable grant object cannot demonstrate that consent was given at a past date: a stream added and later removed leaves no trace, and a grant revoked and re-issued looks identical to one that never lapsed. `ledger.py` records every consent event — granted, stream added, re-affirmed, lapsed, revoked, erased, context declared, refused — as an immutable, hash-chained entry.

Exercised in this pass (`verify_claims.py` §5):

- `streams_granted_at(subject, when)` answers the demonstrability question directly. Queried at the instant a second stream was added it returned the streams then under grant; queried after revocation it returned none.
- `verify()` re-derives the chain and returned `True`. Editing a single historical entry's detail and re-verifying returned `False`. This is tamper *evidence*, not tamper proofing — the subject owns their own file and that is the point — but a controller claiming consent on a date can show a chain that verifies, and an edited one will not.
- **Erasure retains the record, not the data.** The ledger records *that* observations were erased. That record is the evidence the withdrawal was honoured.
- `withdrawal_receipt()` returns what was granted, when it was withdrawn, how many records were erased, and whether the chain still verifies — something the subject keeps. Granting was a guided flow ending in a summary; revocation was a single call ending in silence, and those are not equally easy experiences however few keystrokes each takes.

7.3 The deployment gate

Emotion inference in a workplace or education institution is prohibited under EU AI Act Art. 5(1)(f), applicable since 2 February 2025. A `DeploymentContext` declares where the instance is running — `self`, `consented_individual`, `institutional` or `undeclared` — and gates the streams that supply the biometric basis for affective inference.

Three design decisions are load-bearing, and all three were confirmed by execution in this pass (`verify_claims.py` §4):

Refusal is the default. An `undeclared` context is treated as institutional and refused. An operator who has not said where the system is running has not established that the prohibition does not apply, and the safe reading of silence is the restrictive one. A `self` context is permitted.

The safety exception requires a written basis, not a flag. An empty `safety_exception_basis` still refuses; a non-empty one permits. A boolean can be set by accident or by a config default; a sentence has to be written by someone. The stated basis is recorded and exported with the consent ledger.

The refusal explains itself. The exception message carries the article numbers, the applicability date, the declared context, and the exact call that would claim the exception — because a refusal a developer cannot act on becomes a refusal a developer removes.

A Python module does not make anyone compliant, and the module says so in its own output. What it does is convert a prohibition from a wish into a mechanism.

7.4 The honest enforcement ratio

Of twelve documented prohibitions, six are enforced by a mechanism and covered by a test; six are policy only. The one that moved most recently is the only one currently unlawful to breach in the EU. We suggest that papers proposing ethical architectures should print this ratio. A list of prohibitions communicates intent; the fraction of them a machine will actually refuse communicates the safeguard.

7.5 Data governance

Storage is local and file-based. The subject owns the file, can export it, and can delete it. The package has **zero runtime dependencies**, which is an auditability commitment rather than an aesthetic one: a person who wants to know what the system does can read all of it, and there is no network path in the software.

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8. What the architecture does not buy

On latent-state recovery under model misspecification, the graph-backed inference is **statistically indistinguishable from a median-split quadrant classifier across all 20 tested conditions**, with paired differences of −2.5 to +2.5 points and 2,000-resample percentile bootstrap intervals spanning zero everywhere.

A reviewer may reasonably ask why an interactive-systems venue should care about an architecture that buys no accuracy. The answer is that accuracy was never the safety-relevant quantity: the result of §6.3 depends on the structural commitments — on representing a competing hypothesis rather than filtering, and on distinguishing an absent channel from a zero reading. Structure is what carries it.

We report the negative result prominently because the temptation to omit it is strong and because it relocates the contribution rather than diminishing it. **The architecture is not an accuracy technique.** Its value is structural: conclusions that trace to evidence, provenance that survives derivation, contradiction that is retained, revision that can be undone, inquiries that justify themselves, and conformance that is decidable.

Two related measurements complete the picture.

Calibration. Where the model's assumed structure approximately holds, accumulating evidence with an explicit *none-of-these* component reduces expected calibration error from **0.171 to 0.044** and raises AUC(uncertainty ☐ error) from **0.868 to 0.949**. Where it materially does not hold, calibration is **worse**: ECE 0.151 ☐ 0.221. The architecture trades uniformly mediocre calibration for good-when-applicable and poor-when-not, which is an improvement only to the degree the system can tell which regime it is in. It partly can, and not enough.

Identifiability. A two-channel location model cannot distinguish "state A, displaced" from "state B" when the displacement points toward B. This is not fixable by thresholding. It *is* partially fixable by observing more — pooling across topics to estimate a person-level displacement:

δ	Topics	Uncorrected	Hard-assignment EM	Soft-assignment EM (current)
0.30	5	82.3%	90.8%	**93.4%**
0.30	8	83.8%	95.8%	**98.2%**
0.45	3	54.1%	55.8%	**76.2%**
0.45	5	53.6%	58.4%	**89.5%**
0.45	8	53.2%	61.3%	**97.0%**

A caution that nearly shipped. The first version of this fix made things much worse where the data is thin: at one topic and δ = 0.30, correcting with the unguarded soft estimate drove accuracy from 81.2% to **20.8%**. The cause is identifiability, not tuning — a single point can be moved onto any of four corners by some offset, so there are more unknowns than equations. The estimator now **refuses below three topics** and returns a zero offset. That is a condition, not a threshold. We flag it because a fix that improves a headline by 36 points while quietly destroying the thin-data case is exactly the kind of change that passes a benchmark and fails a person.

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9. Complexity, measured

We report complexity as measurement rather than assertion, as scaling ratio per doubling (linear ≈ 2.0×, quadratic ≈ 4.0×):

Operation	Bound	Measured
Provenance closure `ancestors()`	O(V+E)	**2.0**× — linear
Inquiry planning	O(|H|·|A|)	3.9× at |H|=|A| — product of two linear factors
Belief update, log-odds ledger	O(1) amortised	**4.2× — quadratic ☐ 2.0**× after fix
Write-then-consult path	O(n) per consultation	**3.8× — quadratic ☐ ≈2.0**× after fix

Two rows were defects in our own code. The belief ledger summed every contribution on each call, costing 624 ms at 8,000 updates; maintaining a running total gives 78 ms at 16,000. **A regression test fails if the ratio exceeds 3.0**×, using best-of-three timing so that machine load does not convert a linear algorithm into a failing test.

The fourth row was the larger one. Beliefs are now computed when the mirror is **consulted**, not when an observation is written, which took the path from 1121 ms to 5.24 ms at 800 evidence pairs — a 214× improvement — and from 3.8× to 2.0× per doubling. Re-measuring the write-then-consult path in this pass over 100 ☐ 800 evidence pairs reproduced **linear scaling at roughly 1.8–2.1× per doubling** across repeated runs (`verify_claims.py` §1); absolute timings differ from the figure above because the harness differs, and the shape rather than the constant is the claim.

Two honest notes survive the fix. The reactivity counter now counts consultations rather than writes, which is arguably what a reactivity measure should have counted from the start, since it exists to warn that *looking at a guess changes later answers*. And the asymptotics of a **read-heavy** pattern are unchanged: one consultation per twenty writes still scales at 3.1× per doubling, because each consultation is O(n). The constant improved twentyfold; the shape did not.

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10. Validation

10.1 The test suite

Running the suite in this pass: **707 passed, 2 xfailed**. The two strict `xfail`s record operators that are specified and implemented nowhere, so they cannot quietly become "implemented" in a later summary:

Operator	Status
Causal reflection — intervention, counterfactual query	Prose only. No do-operator exists. Research agenda, not a component
Knowledge consolidation — promotion from belief to revisable knowledge	Prose only. No promotion criterion exists

Also unimplemented: encryption at rest, multimodal acquisition, a reflective execution engine with event queues, and any user interface beyond a command line.

A reference architecture earns the word *reference* when the distance between its formalism and its implementation can be enumerated. We enumerate it.

10.2 How strong the suite actually is

Earlier drafts stated that nothing measured whether the randomised harness exercises the state space in any principled sense. Mutation testing now does, and the measurement is not flattering.

Mutation testing produced **5,114 mutants**. The suite killed **2,319** and **1,549 survived**, for a mutation score of **59.9%**. A further 1,244 mutants sit in code with no covering test at all, 909 of them in a command-line interface that has never had tests and carries no claim in this paper.

Survivors concentrate in the inference engine (404), the pattern layer (278), the invariant checks themselves (206) and the formation model (195). Inspection of a sample shows most are label and message mutations rather than behavioural ones, which means the raw score understates the suite. But not all are cosmetic. **Mutating an invariant's own identifier survived**, which means no test asserted that a check returns the invariant it claims to be — a catalogue of named architectural properties that could have been silently renamed. Two tests now close that, and they exist because a tool found the gap rather than because we thought of it.

We report the score rather than tuning it. A mutation score is a lower bound on test strength and an upper bound on nothing, and 59.9% is a middling result for a package that makes conformance claims.

10.3 Five things found by measurement rather than by intent

Listed because they are the best argument for the method, and because each was slightly embarrassing.

1. **Provenance as a scalar dropped every parent but one.** The formalism said "attribute" where the literature had said "set" three decades earlier. Invariant II caught it.
2. **A clamping step that looked like numerical hygiene destroyed commutativity and retractability.** Belief order-invariance went from 0.0396/0.9900 across permutations to exact agreement.

3. **Invariant X false-positived on our own shipped output.** The defect was in the check, not the output (§5.3).
4. **Retroactive consent was already impossible and nobody designed it.** Found by a failing test. It falls out of asking *what was granted at time T* rather than *is there a grant*.
5. **Mutating an invariant's own identifier survived.** Nothing asserted that a check returns the invariant it claims to be (§10.2).

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11. The regulatory tension

We report this because it is unresolved, and because the shape of it is more general than this system.

The EU AI Act defines biometric data as *"personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person"* (Art. 3(34)). Unlike GDPR Art. 4(14), **it does not require unique identification**. A heart-rate-like arousal proxy is therefore biometric data under the Act while identifying nobody. An *emotion recognition system* is one *"for the purpose of identifying or inferring emotions or intentions of natural persons on the basis of their biometric data"* (Art. 3(39)). Our guard infers a threat response — an affective and intentional state — partly on the basis of that arousal channel.

So: **the channel that exists only to protect people, and that §6.4 shows does protect them, is what brings the system within a prohibition written to prevent a different harm.** Remove it and the exposure largely disappears; remove it and the false hidden-desire rate returns to 31.2%.

The safety exception at Art. 5(1)(f) is available and, in our reading, applicable: the channel's sole documented purpose in code, tests and specification is to prevent a specific measured harm, and it is used for nothing else. We state that as an argument rather than a conclusion. We are aware of no case law interpreting the exception for a guard of this kind, and a system whose safety mechanism is itself an emotion-inference system is a case the Act does not obviously anticipate.

11.1 A second regime, with no research exemption

The EU analysis is the one the literature rehearses. It is not the only regime a system like this must survive, and the contrast is instructive.

This work is conducted under Egyptian jurisdiction. Law No. 214 of 2020 on Clinical Medical Research establishes a national ethics architecture but does not reach us: its subject is defined as studies assessing "therapeutic, pharmaceutical, surgical, nutritional, preventive or diagnostic interventions", and behavioural, psychological and computing research falls outside its scope. The Supreme Council of Universities issued a national academic-integrity and research-ethics guide in January 2026 whose third book addresses behavioural and social research committees, but it carries no decree number and no binding-force clause, and whether it reaches private universities is unspecified. We classify it as guidance rather than law.

Personal Data Protection Law No. 151 of 2020 does reach us, and reaches further than the GDPR in three respects. Its definition of personal data expressly includes data determining "psychological … identity". Its sensitive-data category covers "psychological, mental, or physical health, or genetic, biometric or financial data", and Article 12 prohibits processing that category without a licence from the Personal Data Protection Centre *in addition to* written explicit consent, on penalty of imprisonment and a fine up to EGP 5,000,000. And — the point worth putting in front of any venue — **it contains no scientific-research exemption whatsoever**. The GDPR's research accommodations have no Egyptian counterpart. A search of the Arabic text of both the law and its executive regulations (Decree No. 816 of 2025, in force 2 November 2025, compliance deadline 31 October 2026) finds that the words for "research", "studies" and "university" do not appear.

The consequence inverts a common assumption. A researcher in a jurisdiction with a *weaker* national human-subjects law can face a *stricter* data-protection constraint on exactly the physiological and psychological channels this architecture consumes, with no research carve-out to fall back on. Our design response is the one already described: refuse the biometric stream rather than seek a basis for it, and treat anonymisation rather than pseudonymisation as the only route out of scope.

This is a reading of statutory text, not legal advice, and it has not been reviewed by counsel.

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12. Limitations

No human subjects. Nothing here establishes any effect on human reasoning, comprehension, trust, wellbeing or task outcome, and no such claim is made. Every result is a property of code or of simulated signal streams.

Simulation encodes a theory. The generative processes embody assumptions about how latent states manifest in observable signal. If those are wrong, absolute rates are wrong in the same direction for every method compared, including the baselines. We regard the *ordering* of methods as more robust than the absolute rates, and §6.7 measures how far that ordering survives as the key assumption is weakened.

A point estimate of zero is not a guarantee of zero. Across 800 threat episodes we observed no misreadings. The 95% upper bound implied by the rule of three is **0.37%**, and that — not 0.0% — is the number a deployment decision should use.

Two implementations, not many. §5.4 shows the decision procedure separating a compliant implementation from a non-compliant one, which rules out vacuity. It does not establish that the procedure is portable to an implementation built on different primitives — a graph database, an event-sourced store, a logic programming substrate.

Invariants X and XI are syntactic proxies for semantic properties. Whether an output usurps a person's authority over their own self-understanding is not a matter of which pronouns it contains. Our check tests phrasing, and §5.3 documents it producing a false positive on its first contact with real output. It will also produce false negatives: a sentence can be a verdict without matching any pattern we listed. The check raises the floor; it does not measure the property.

The clinical vocabulary is borrowed. We use "compliance" and "threat response" to name a pattern of approach behaviour under low stated desire and high arousal. There is a genuine peer-reviewed literature in this region — appeasement under traumatic entrapment, tonic immobility as an evolved defence, the acute stress response as a spectrum — and we cite it to locate the concern, not to claim our simulation instantiates any of it. One term deserves separate treatment: **"fawn" is not a clinical category.** We traced it to a self-published practitioner trade book and could locate no refereed source establishing it as a validated construct. It has been removed from every load-bearing claim.

We are not lawyers. §11 is a reading of statutory text, not advice.

The specificity cost is real. Under partial observation the system labels excitement as a possible threat 99.0% of the time. A person who is excited and told to consider whether they are afraid has been mildly misread. We judge this preferable to the converse and do not claim it is costless.

Credible sets do not reach nominal coverage. Reporting a credible set instead of a point label improves the rate at which the truth is contained in the output (78.2% □ 88.0% at $\delta = 0.30$) but a set claiming 90% mass contains the truth 60.5% [55.5, 65.2] of the time at $\delta = 0.45$. A credible set inherits its posterior's miscalibration; reaching nominal coverage requires conformal calibration against held-out labelled outcomes, which do not exist without a human study.

Scale. Everything reported runs on a single machine over thousands of nodes, not millions, and the read path remains O(n) per consultation.

13. Conclusion

Making the invariants executable found two defects that a prose specification had asserted away: a provenance model that dropped parents, and a belief operator whose result depended on the order evidence arrived. Neither was visible to a reader of the document. Both were visible on the first run of the decision procedure.

Making refusal a first-class output, rather than a low-confidence label, removed a specific foreseeable harm in simulation — and the ablation shows that the removal depends on a commitment that looks like a detail: an absent channel must be representable as absent. A system that imputes it to zero keeps its guard in the codebase and loses it in production, with nothing in the output to say so.

We think the transferable claims are these. Reference architectures for human–AI interaction should ship their conformance suites, should include negative tests, and should point those suites at output a person would actually see. Papers proposing ethical architectures should print the fraction of their prohibitions that a machine will actually refuse. And a reflective system under partial observation should refuse rather than degrade.

Ours did all three and was wrong three times — about provenance, about order invariance, and about its own Invariant X check. Each error surfaced within a minute of the check being run, and none would have surfaced from reading.

Ethics statement

This research did not involve human or animal subjects. All results derive from code execution and from simulated signal streams; no personal data was collected, processed or analysed.

The system is a research instrument, not a medical device, and is not suitable for clinical, educational, employment, credit, insurance, policing or other high-stakes deployment. It refuses to enable the physiological stream in institutional and undeclared deployment contexts. The authors note the regulatory position described in §11 and do not assert that the safety exception under Art. 5(1)(f) applies as a settled matter.

Data and code availability. Implementation and conformance suite are released under an open licence. *[REPOSITORY URL — TO BE COMPLETED]* See Appendix A for the provenance of every number in this paper, including which were re-derived in preparing this draft and which were not.

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The 102 references below are the union of the two source papers' reference lists, deduplicated. Each was confirmed against a live source with a resolvable identifier in Session 14, **except the 6 marked** †, which were introduced in Session 15 after the verification pass and are unconfirmed under ground rule 6; see **OW-9** in Appendix B.

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Appendix A — Number provenance

Ground rule 4 of this programme: *no number enters a manuscript unless a file in the repository produces it and the person writing the sentence has run that file.* This appendix records compliance with that rule honestly, which means recording where it could not be met.

Re-derived while preparing this draft, by running the named file against `shadow_mirror` v1.3.0:

Claim	§	Source	Result
Test suite totals	10.1	`pytest -q`	707 passed, 2 xfailed
Order invariance over 120 permutations	4.2, 5.4	`verify_claims.py` §2	120 permutations, 1 distinct outcome, spread 0.0
Eleven callable invariant checks	5.1	`verify_claims.py` §3	11 checks + `check_all`
Deployment gate refuses institutional and undeclared	7.3	`verify_claims.py` §4	refused; `self` permitted
Empty safety basis still refuses	7.3	`verify_claims.py` §4	`True`
Ledger verifies; edit is detectable	7.2	`verify_claims.py` §5	`verify()` `True` ☐ `False` after edit
`streams_granted_at` is time-aware	7.2	`verify_claims.py` §5	returns granted streams, then none after revocation
Refusing methods refuse when called	5.5	`verify_claims.py` §6	9 called, 9 refused, 0 returned

Claim	§	Source	Result
Write-then-consult path is linear	9	`verify_claims.py` §1	≈1.8–2.1× per doubling

Carried from the programme record, and *not* re-derived here. The scripts that produce these numbers — `run_benchmarks.py`, `conformance_v10.py`, `experiment_d23.py`, `experiment_ambivalence.py`, `analyze_v030_vs_v040.py` — and their outputs `results.json` and `v030_vs_v040.json` are **not present in the research archive this draft was assembled from**. They are cited here on the authority of the session reports and the two source papers, and a reader cannot currently re-run them. This is recorded as **OW-6** in Appendix B and is, in our judgement, the most serious reproducibility defect in the present state of the work.

Claim	§	Cited source
All safety rates, intervals and per-process figures	6.3	`results.json → b2.conditions[1]`
Ablation figures	6.4	`results.json → b2.conditions[2]`
Refusal calibration	6.5	`results.json → b3.results`
Ambivalence, pre- and post-fix	6.6	`experiment_ambivalence.json`; pre-fix recorded in Session 8
Separation sweep and supervised baseline	6.7	Session 13 adversarial review
Inquiry regret	4.3	`results.json → b4.conditions`
Median-split parity across 20 conditions	8	`results.json → b1.cells`
Calibration ECE and AUC	8	`v030_vs_v040.json`
Identifiability and offset pooling	8	`experiment_d23.json`; soft-EM figures from Session 12
Belief-ledger and prototype complexity ratios	9	`test_operator_correspondence.py` timing harness
Mutation score and survivor distribution	10.2	Session 17 mutation run
Prototype conformance 4/7	5.4	`conformance_v10.py`

Reconciliation note. Where the two source papers and the current package disagree, this draft follows the package and says so. Four figures were superseded after the papers were drafted: the write-then-consult path is no longer quadratic (§9); the identifiability estimator now uses soft assignment and refuses below three topics (§8); the consent ledger and withdrawal receipt now exist (§7.2); and the deployment gate now exists (§7.3), moving the enforcement ratio from five of twelve to **six of twelve** (§7.4). The papers in `papers/` still carry the pre-Session-12 values. This is recorded as **OW-7** in Appendix B.

To reproduce what can be reproduced:

```
pytest -q                # 707 passed, 2 xfailed
python verify_claims.py  # writes verify_claims_output.json
python demo.py
```

Appendix B — Outstanding work

This appendix is deliberately separate, and it is the appendix a reader should check before believing anything above.

It has three parts. **B.1** lists what has no available substitute — the rule applied is that an item appears only if an active hunt for a public dataset, a free instrument, a third-party service or a methodological substitute came back empty. **B.2** lists what a search *resolved*, so that B.1 reads as the residue of a search rather than as its whole. **B.3** lists ordinary unfinished work, which is excluded from B.1 on purpose because none of it is

blocked on anything external.

B.1 Blocked: no available substitute

OW-1 · The threat-confound claim requires primary data collection. Approximately forty public datasets were assessed across affective computing, approach-avoidance paradigms, ecological momentary assessment, experimental economics and social conformity. The claim needs four things measured on the same person and the same item: a stated stance, a behavioural choice that can contradict it, an arousal channel time-locked to that choice, and a coercion manipulation. That intersection does not exist publicly. The failure is structural rather than incidental — corpora with good physiology are passive-viewing paradigms in which nothing can contradict a rating, and corpora with real choice lack autonomic recording, self-report, or both. Two published studies used almost exactly the right design and deposited no data. **No workaround exists. Only collection.**

OW-2 · No instrument in the battery is validated in an Egyptian sample. Arabic validations were confirmed for Lebanese, Saudi, Omani, Jordanian, Palestinian, Qatari and Moroccan samples. None is Egyptian. Four candidate Egyptian-journal papers were identified by DOI, but every `*.ekb.eg` host was unreachable, so their contents are unknown — and Egyptian psychometric papers frequently construct local instruments rather than translating Western ones, so their relevance cannot be assumed. Any use of this battery in Egypt is first-in-population work and must be described as such. **Partially resolvable by a colleague with Egyptian institutional access; not resolvable by open search.**

OW-3 · Two instrument constructs have no free replacement. There is no openly licensed eight-theme Eriksonian inventory: the two structurally correct candidates are both permission-gated, and going upstream from one to the other does not escape the gate. The best free material covers two of eight themes and yields single scores rather than subscales. For purpose beyond the Meaning in Life Questionnaire, no free substitute was identified. **Resolvable only by author permission, which is a letter, not a search.**

OW-4 · No party outside this project has executed the software. This was the largest risk identified by the programme's own adversarial review. The venue route is closed: neither ACM TiiS, nor ACM FAccT, nor AAAI/ACM AIES operates an artifact-evaluation or badging track an author can enter at submission time. Archival deposit satisfies an availability criterion but verifies no execution. Two replication communities publish work written *by* the replicator and therefore cannot be pointed at an artifact. One community code-check service matches the need exactly and has **not been used**. The claim in this work is that the code is inspectable, not that it has been independently run.

OW-5 · Invariants X and XI are syntactic proxies for semantic properties. Invariant X detects verdict phrasing by pattern, and the question generator is guarded against prescriptive drift by a keyword test. Both raise a floor and neither measures the property. We know of no decision procedure that settles whether a sentence claims authority over a person, and we do not think one follows from more patterns. This is a research problem rather than an unfinished task.

B.2 Newly identified while preparing this draft

OW-6 · The benchmark scripts and their outputs are not in the archive. [blocking for submission] Both source papers state that every number is produced by a named script a reviewer can run. As assembled, the archive contains the package, the papers, the session reports and the submission material — but **not** `run_benchmarks.py`, `conformance_v10.py`, `experiment_d23.py`, `experiment_ambivalence.py`, `analyze_v030_vs_v040.py`, `make_paper_figures.py`, `results.json` or `v030_vs_v040.json`. Every headline safety figure in §6, the calibration and identifiability figures in §8, and the conformance comparison in §5.4 therefore cannot be re-derived from what is on hand. The figures under `papers/figures/` exist but their generator does not. **Action:** locate these files in the working environment that produced them and commit them, or regenerate and re-verify every number they support before submission. Until then the availability statement in the Ethics section overclaims.

OW-7 · The two source papers carry pre-Session-12 figures. [blocking for submission] P1 §7 still reports the write-then-consult path as quadratic (3.8× per doubling, 1121 ms at 800 pairs); P2 §9 repeats it. P1 §8 reports identifiability at $\delta = 0.45$ with eight topics as 61.3%, from the superseded hard-assignment estimator. P2 §8 reports that no append-only consent record exists and that no deployment-context mechanism has been implemented, and prints an enforcement ratio of five of twelve. All four were superseded by Session 12 and are corrected in this unified draft. **Action:** propagate the corrections into `P1_ACM_TiiS_draft.md` and `P2_AIES_FAccT_draft.md`, or withdraw those files in favour of this one. Submitting either paper as it stands would publish four figures the authors know to be stale.

OW-8 · Test-count drift across documents. The archive README reports 707 passing; P1 reports 673 collected and 671 passing with a 563-invariant breakdown; Session 12 reports 324. The first is correct for v1.3.0 and was reproduced here. The per-category breakdown in P1 §3.6 has not been recomputed for the current suite, so this draft does not restate it. **Action:** recompute the breakdown from `pytest --collect-only` and state it once, in one place.

OW-9 · Nine reference-list entries sit outside the verified set. `bibliography.json` records 105 manuscript references verified in Session 14 and reports `entries_without_resolvable_identifier: 0`. The papers' reference lists are longer: P1 lists 58 and 55 are in the verified set; P2 lists 56 and 50 are. The nine remaining entries are six unique psychometric instrument sources, all introduced in Session 15 — after the verification pass — and never folded into it. They are marked † in the reference list above. This is not a claim that any is wrong, but under ground rule 6 they are unconfirmed. **Action:** verify the six and update `counts`, or mark them explicitly as pending.

OW-10 · Two release decisions the repository now forces. Neither is a research defect; both follow from the archive having been committed. First, the full working conversation transcript is in the repository under `archive/`. The repository is private, so this is storage rather than publication — but the data-availability statement above promises a *public* repository, and the transcript would go public with it. Second, the v1.3.0 archive currently sits on a branch rather than on `main`, so a repository URL quoted today would resolve to the superseded v0.3.0 package. **Action:** decide deliberately whether the transcript is published, moved to a controlled deposit, or replaced by a code-only mirror; and merge the branch before quoting a URL.

B.3 Ordinary unfinished work

Excluded from B.1 on purpose: none of it is blocked on anything external, and listing it there would dilute a list whose value is that every entry has been searched for and not found.

Item	Status
Causal reflection operator	Declared gap, strict `xfail`. No do-operator exists
Knowledge consolidation operator	Declared gap, strict `xfail`. No promotion criterion exists
Read path still O(n) per consultation	Open. Anchored-decay accumulators would fix it

Item	Status
1,549 surviving mutants	Open. Concentrated in engine, patterns, invariant checks, formation
909 mutants in an untested CLI	Open. Carries no claim in any paper
Candidate belief state for monotonic drift	Proposal ("your position moved steadily"), computed and discarded
Conformal calibration for credible sets	Blocked on labelled outcomes, i.e. on a human study
Encryption at rest, multimodal acquisition, event-queue execution engine, any GUI	Specified, unimplemented

B.4 Before submission

Item	Who can close it
Corresponding email, and a department if one applies	**You.** The only gap in the author block
Repository URL in both availability statements	**You**
Venue selection for the ethics paper (AIES or FAccT)	**You.** Check current calls; dates move annually
OW-6 and OW-7 above	Whoever holds the working environment that produced `results.json`
Three dataset downloads for the secondary-analysis route	Mechanical; two are small
Urban Appraisal rating polarity	From the BeMoBIL analysis code. The loader refuses until supplied
Two instrument permissions (Claremont Purpose Scale; MEPSI)	Letters to the instrument authors
CITI certification and a supervisor	Only if the primary study proceeds

B.5 What would change the conclusions

Stated so that a reader knows what evidence would move this work rather than merely extend it.

1. **A human-subjects study.** Objective O4 is unanswered. Every claim about reflection, comprehension or trust is currently a design intention with no evidence, and a null result would be a finding.
2. **Primary collection on the four-way intersection (OW-1).** The safety result is a property of a simulator that encodes our own theory. Real data could refute the ordering of methods, not merely its magnitude.
3. **A third party running the conformance suite against a different substrate (OW-4).** Until then, "decidable" is demonstrated and "decidable usefully" is not.
4. **A decision procedure for semantic authority (OW-5).** Invariants X and XI currently raise a floor they cannot measure.